Advanced Data Processing for Retrieval

This document exercises the parsing pipeline for complex documents. It contains a nested heading hierarchy, a real table drawn with vector lines, and a vector figure. None of these carry PDF semantics, so a parser can only recover them from geometry.

1. Why flat text extraction fails

A flat text extractor reads the drawing operators in order and treats every glyph run as a paragraph. Tables therefore collapse into a run of values with no row or column meaning, and figure labels end up mixed into the surrounding prose. Retrieval on top of such text cannot answer questions that require reading across a row.

2. Measured layout drift

The table below compares three extraction strategies on the same document. The column order reflects the reading order the parser recovered, not the logical order of the source.

Strategy	Tables	Reading order	Cost
Flat text	none	column major	low
Column aware	partial	row major	medium
Layout model	full	row major	high

Figure 1 shows the drift of the recovered reading order as the document gets denser. The horizontal axis is the number of regions on a page and the vertical axis is the fraction of regions that came back in the wrong order.

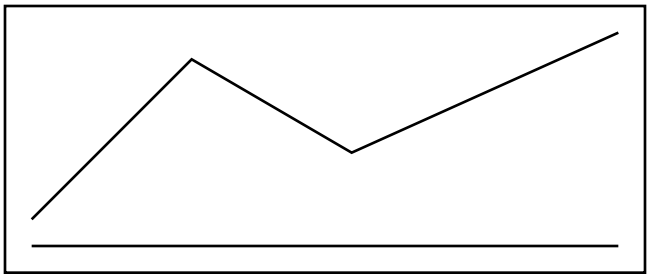

Figure 1. Reading order drift against region density.

3. What a table has to preserve

A table has to preserve three things at once: the header row, the row boundaries, and the cell to column mapping. Losing any one of them makes the table worse than useless, because it looks structured to a reader while carrying no recoverable meaning for a retriever.

Property	Why it matters	Loss symptom
Header row	Names the columns	Values lose their axis
Row boundary	Groups related cells	Rows merge into prose
Column map	Enables cell lookup	Cross row answers fail

The last paragraph is deliberately plain prose so that a parser which only handles tables has something to fail on.